In [1]:
```
import pandas as pd
import numpy as np

df = pd.read_csv("RealEstate_Prices.csv")

print(df.head())
print(df.info())
print(df.isnull().sum())

df.columns = (
    df.columns
    .str.strip()
    .str.lower()
    .str.replace(" ", "_")
    .str.replace(r"[^a-zA-Z0-9_]", "", regex=True)
)

df = df.drop_duplicates()

for col in df.select_dtypes("object"):
    df[col] = df[col].str.strip()

df = df.dropna()

categorical = df.select_dtypes("object").columns
df = pd.get_dummies(df, columns=categorical, drop_first=True)

numeric = df.select_dtypes(include=np.number).columns

for col in numeric:
    q1 = df[col].quantile(0.25)
    q3 = df[col].quantile(0.75)
    iqr = q3 - q1
    df = df[
        (df[col] >= q1 - 1.5 * iqr) &
        (df[col] <= q3 + 1.5 * iqr)
    ]

print("\nCleaned Data:")
print(df.head())

print("\nAverage values:")
print(df.mean(numeric_only=True))

df.to_csv("cleaned_realestate_prices.csv", index=False)
```

```
      price  area  bedrooms  bathrooms  stories mainroad guestroom basement  \
0  13300000  7420         4          2        3      yes        no       no
1  12250000  8960         4          4        4      yes        no       no
2  12250000  9960         3          2        2      yes        no      yes
3  12215000  7500         4          2        2      yes        no      yes
4  11410000  7420         4          1        2      yes       yes      yes

  hotwaterheating airconditioning  parking prefarea furnishingstatus
0              no             yes        2      yes        furnished
1              no             yes        3       no        furnished
2              no              no        2      yes   semi-furnished
3              no             yes        3      yes        furnished
4              no             yes        2       no        furnished
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 545 entries, 0 to 544
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   price             545 non-null    int64
 1   area              545 non-null    int64
 2   bedrooms          545 non-null    int64
 3   bathrooms         545 non-null    int64
 4   stories           545 non-null    int64
 5   mainroad          545 non-null    object
 6   guestroom         545 non-null    object
 7   basement          545 non-null    object
 8   hotwaterheating   545 non-null    object
 9   airconditioning   545 non-null    object
 10  parking           545 non-null    int64
 11  prefarea          545 non-null    object
 12  furnishingstatus  545 non-null    object
dtypes: int64(6), object(7)
memory usage: 55.5+ KB
None
price               0
area                0
bedrooms            0
bathrooms           0
stories             0
mainroad            0
guestroom           0
basement            0
hotwaterheating     0
airconditioning     0
parking             0
prefarea            0
furnishingstatus    0
dtype: int64

Cleaned Data:
      price  area  bedrooms  bathrooms  stories  parking  mainroad_yes  \
15  9100000  6000         4          1        2        2          True
20  8750000  4320         3          1        2        2          True
22  8645000  8050         3          1        1        1          True
27  8400000  8875         3          1        1        1          True
40  7875000  6550         3          1        2        0          True

    guestroom_yes  basement_yes  hotwaterheating_yes  airconditioning_yes  \
15          False          True                False                False
20          False          True                 True                False
```

```
22           True          True                False                True
27          False         False                False               False
40          False          True                False                True

    prefarea_yes  furnishingstatus_semi-furnished  \
15         False                             True
20         False                             True
22         False                            False
27         False                             True
40          True                            False

    furnishingstatus_unfurnished
15                         False
20                         False
22                         False
27                         False
40                         False

Average values:
price                             4.045012e+06
area                              4.623811e+03
bedrooms                          2.742466e+00
bathrooms                         1.000000e+00
stories                           1.553425e+00
parking                           5.369863e-01
mainroad_yes                      8.383562e-01
guestroom_yes                     1.397260e-01
basement_yes                      3.178082e-01
hotwaterheating_yes               3.835616e-02
airconditioning_yes               2.356164e-01
prefarea_yes                      2.082192e-01
furnishingstatus_semi-furnished   4.109589e-01
furnishingstatus_unfurnished      3.780822e-01
dtype: float64
```